TokenBridge

Business Requirements Document

Cross-Border Tokenized Settlement Platform*Cross-Border Tokenized Settlement Platform*

Prepared by: Devender | Senior Business Analyst

1. Executive summary

TokenBridge is a portfolio project that demonstrates a tokenized alternative to the traditional correspondent banking model for cross-border payments. It applies patterns from real institutional tokenization initiatives (such as Project Guardian) to a smaller, demonstrable scope: two tokenized currencies settling an FX trade through an atomic delivery-versus-payment (DvP) mechanism, with compliance enforced at the smart contract level rather than only in application logic.

This document captures the business problem, stakeholders, as-is and to-be process flows, solution architecture, and the functional and non-functional requirements that the accompanying product backlog is built from.

2. Business problem

Cross-border payments today typically move through a chain of correspondent banks. Each hop adds cost, delay (often 2-5 business days), manual reconciliation effort, and a settlement risk window where one party has paid but not yet received. Liquidity is trapped in pre-funded nostro/vostro accounts to cover this delay.

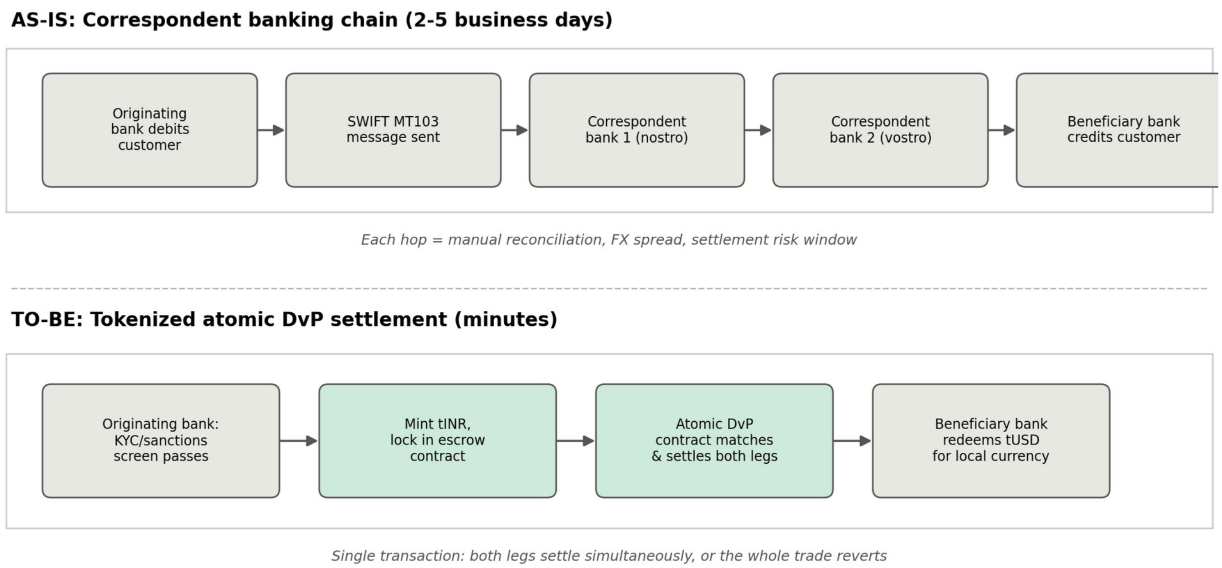
Regulators and central banks (including through initiatives like Project Guardian) have been piloting tokenized deposits and atomic settlement to remove this risk window and unlock trapped liquidity, while keeping compliance controls intact.

3. Stakeholders

Stakeholder	Interest / role
Originating bank treasury ops	Initiates cross-border payment, needs speed and cost certainty
Beneficiary bank	Receives and credits funds, needs settlement finality guarantee
Compliance / MLRO	Ensures KYC, sanctions screening, and audit trail on every transfer
Regulator / auditor	Needs read-only visibility into settlement and compliance events
Treasury / liquidity desk	Manages FX rate exposure and nostro/vostro funding

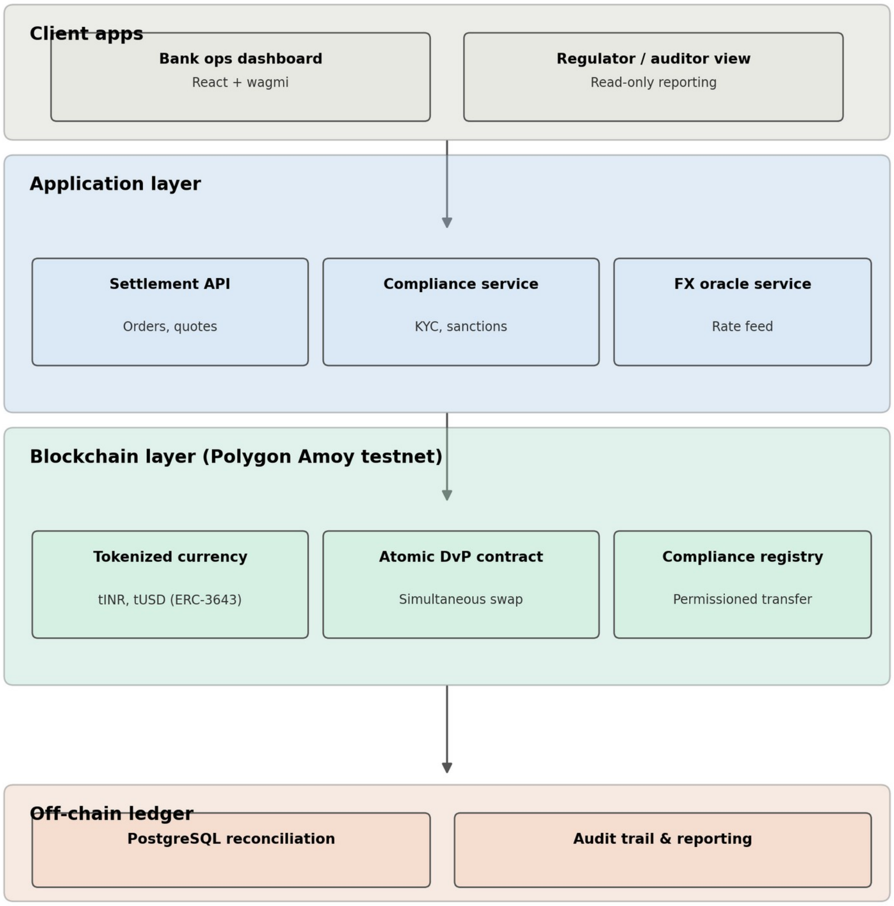
4. As-is vs to-be process

The as-is flow relies on a chain of correspondent banks passing SWIFT messages; the to-be flow settles both legs of the trade in a single atomic transaction.



5. Solution architecture

Four layers: client applications, an application/API layer handling compliance and FX rates, a blockchain layer holding the tokenized currency and settlement logic, and an off-chain ledger for reconciliation and audit.



6. Scope

In scope

- Tokenized representations of two currencies (tINR, tUSD) on a public testnet
- Atomic DvP settlement contract
- Permissioned compliance registry (mock KYC/sanctions allow-list)
- FX rate oracle (simulated feed)
- Ops dashboard for initiating and monitoring settlements
- Off-chain reconciliation ledger and audit trail

Out of scope

- Real fiat on/off ramp or custody of real funds
- Integration with live SWIFT or a real core banking system
- Production-grade key management / HSM integration

7. Non-functional requirements

Category	Requirement

Performance	Settlement finality under 2 minutes on testnet
Security	Only allow-listed wallets can hold or transfer tokenized currency
Auditability	Every on-chain event mirrored to an immutable off-chain audit log
Availability	API layer target 99.5% uptime in demo environment
Compliance	Transfers blocked at contract level if sender/receiver fails screening

8. Compliance and risk framework

- Every wallet must be allow-listed by the compliance registry contract before it can hold or receive tokenized currency
- Sanctions/KYC screening happens off-chain first (compliance service); the result is what adds/removes an address from the on-chain allow-list
- Atomicity removes principal (Herstatt) risk: both legs settle together or the transaction reverts entirely
- Every on-chain transfer emits an event that is mirrored into the off-chain audit log for regulator reporting

9. Assumptions and constraints

- Deployed to a public EVM testnet (Polygon Amoy) rather than a private/permissioned chain, for demonstrability
- FX rates are simulated rather than sourced from a live market feed
- Single demo bank pair (originating / beneficiary) rather than a multi-bank network

10. Glossary

- DvP (Delivery versus Payment): settlement method where transfer of an asset and payment happen simultaneously
- Atomicity: a transaction either completes fully or has no effect at all
- Nostro/vostro account: accounts banks hold with each other to settle cross-border transactions
- Allow-list / permissioned token: a token that only permits transfers between pre-approved addresses